

6G Weekly Survey — Jun 22 - Jun 28, 2026

24 items across 4 topics, ranked by importance.

Channel modeling / GAN-based

Point-Cloud-Assistant Localized Statistical Channel Prediction by Tangent Gaussian Splatting

Ye Xue, Yiheng Wang, Xinhua Shao, Qi Yan, et al. · arXiv eess.SP · 2026-06-17

★★★★■ Strong contribution bridging ray-tracing-free statistical channel modeling with geometric priors from digital twins, validated on realistic city-scale data and directly applicable to 6G network planning with sparse measurements. The paper addresses the inability of localized statistical channel models (LSCM) to predict angular power spectrum (APS) at unmeasured locations by proposing PC-TGS, which fuses sparse RSRP measurements with dense LiDAR point clouds. Environmental scatterers are represented as anisotropic 3D Gaussians initialized from point clouds, then projected to local angular domains and aggregated via depth-aware electromagnetic splatting. Evaluation on a city-scale dataset (5M LiDAR points, 6,310 RSRP samples) shows improved APS/RSRP prediction over baselines with faster inference for spatial extrapolation.

Contribution: First framework to extrapolate angular power spectrum to unmeasured locations by representing scatterers as learnable 3D Gaussians initialized from LiDAR geometry, with provable closed-form integration and electromagnetic splatting.

Relevance to your work: Highly relevant to geometry-aided channel modeling and digital-twin workflows for beyond-5G deployment, offering a computationally efficient alternative to full ray-tracing while leveraging LiDAR assets already used in automotive and urban sensing.

ISAC (sensing + communication)

Hybrid TRP-UE Sensing for Enhanced Target Localization

Necati Kagan Erkek, Marco Di Renzo, Arman Shojaeifard, Yasser Mestrah, et al. · arXiv cs.IT · 2026-06-18

★★★★■ Addresses a critical ISAC deployment challenge with a practical hybrid architecture aligned with 3GPP standardization efforts, validated on compliant channel models relevant to Release 19/20 sensing work items. This paper proposes a hybrid sensing architecture combining network-side (TRP) and user-equipment-side (UE) measurements to improve passive target localization in ISAC systems. The approach is evaluated using 3GPP-compliant channel models for challenging scenarios like indoor factory environments. Results show that UE-assisted sensing significantly enhances localization accuracy compared to TRP-only sensing in NLOS-rich propagation conditions.

Contribution: First demonstration of hybrid TRP-UE collaborative sensing for passive target localization using 3GPP-standardized ISAC channel models, showing practical benefits in industrial indoor scenarios.

Relevance to your work: Directly relevant to your ISAC research interests, particularly for understanding how network and UE cooperation can overcome challenging propagation environments in practical sensing deployments.

Cell-Free Integrated Sensing and Communication

Diluka Galappaththige, Chintha Tellambura · arXiv eess.SP · 2026-06-16

★★★★■ Timely synthesis from credible authors bridging two active 6G research areas (ISAC and cell-free), providing architectural foundation as distributed ISAC emerges in O-RAN and beyond-5G standardization.

This monograph surveys cell-free integrated sensing and communication (CF-ISAC), which combines distributed access-point architectures with joint radar and communication capabilities. It covers fundamentals including multi-static sensing advantages, cooperative transmission schemes, target parameter estimation,

and resource allocation strategies. The work addresses practical challenges like synchronization across distributed nodes, multi-target detection, fronthaul constraints, and interference management in CF-ISAC systems.

Contribution: First comprehensive survey consolidating CF-ISAC principles, distinguishing it from cellular ISAC through distributed multi-static sensing gains and cell-boundary-free resource optimization.

Relevance to your work: Directly relevant as cell-free architectures offer unique sensing geometry and CSI diversity for your ISAC work, while distributed coordination challenges align with your AI-native PHY interests for joint beamforming and resource allocation.

Auto-correlation Function Keying

Weijiang Zhao, Yifeng Xiong, Fan Liu, Shuangyang Li, et al. · arXiv cs.IT · 2026-06-16

★★★★■ Strong theoretical foundation with asymptotic optimality proof and practical transceiver design for communication-centric ISAC, addressing a real limitation (spurious sidelobe peaks) in dual-function radar-communication systems relevant to 6G.

This paper proposes auto-correlation function keying (ACFK), a modulation scheme for communication-centric ISAC that embeds data symbols directly onto ACF-domain sidelobes to achieve strict peak sidelobe level (PSL) control for weak-target detection. The authors formulate mutual information maximization under PSL constraints, prove ACF-domain uniform construction is asymptotically optimal at high SNR, and provide BER analysis with non-negativity violation bounds. Numerical results demonstrate superior PSL control and weak-target detection compared to probabilistic amplitude shaping baselines.

Contribution: First modulation scheme to enable exact control of nominal periodic auto-correlation function by direct ACF-domain symbol embedding, addressing the critical gap between expected sidelobe level (ESL) and peak sidelobe level (PSL) for weak-target detection in ISAC.

Relevance to your work: Directly relevant to integrated sensing and communication (ISAC/JCAS) research, particularly for waveform design that balances communication performance with sensing requirements for weak-target detection in shared-spectrum systems.

BASIIS: Bistatic Angular Sampling and Interpolation for ISAC Setups

Alexander Felix, Marcus Henninger, Lucas Giroto, Maximilian Bauhofer, et al. · arXiv eess.SP · 2026-06-16

★★★★■ Strong theoretical contribution addressing a fundamental efficiency bottleneck in bistatic ISAC with rigorous coarray analysis and significant (3-5×) overhead reduction validated experimentally.

This paper addresses the high overhead of bistatic ISAC by proposing a minimal angular sampling framework that exploits the coupling between transmit and receive elevation angles. The authors introduce the ortho-baseline coarray concept to capture joint elevation aperture and derive a DFT-based sampling and interpolation scheme that reduces required TX-RX direction pairs by 3-5× while maintaining near-lossless target detection accuracy. Monte Carlo simulations validate that minimal sampling achieves detection performance equivalent to dense oversampled imaging.

Contribution: First complete angular sampling framework for bistatic ISAC that accounts for coupled TX-RX elevation geometry through the ortho-baseline coarray, enabling provably minimal sampling in the four-dimensional angle-of-departure/angle-of-arrival space.

Relevance to your work: Directly relevant to your ISAC research interests; this sampling framework reduces radio resource overhead for bistatic sensing operations and could inform beamforming strategies for massive MIMO ISAC deployments where angle-domain efficiency is critical.

Joint Synchronization and Radar Parameter Estimation for Distributed OFDM-ISAC Systems

Niclas Führling, Hyeon Seok Rou, Kuranage Roche Rayan Ranasinghe, Giuseppe Thadeu Freitas de Abreu, et al. · arXiv eess.SP · 2026-06-16

★★★★■ Addresses a fundamental challenge in distributed ISAC architectures with a principled solution backed by CRLB-optimal performance, directly relevant to emerging 6G JCAS system design.

The paper addresses synchronization in distributed ISAC systems operating over doubly-dispersive channels by formulating a joint estimation problem for time/frequency offsets at multiple base stations and target delay/Doppler parameters. A bivariate Gaussian belief propagation algorithm exploits the linearized OFDM-ISAC system model to simultaneously solve for synchronization and radar parameters. Simulations show CRLB-achieving performance at low-to-moderate SNR for both synchronization (TO/CFO) and sensing (range/velocity) estimates.

Contribution: First joint framework treating distributed BS synchronization and radar parameter estimation as a unified inference problem solved via belief propagation on the OFDM-ISAC signal structure.

Relevance to your work: Highly relevant—combines ISAC with OFDM waveforms and tackles distributed synchronization, a critical enabler for multi-BS joint sensing scenarios in your Beyond-5G research portfolio.

Amplitude-Phase-Frequency Block Modulation for OFDM-ISAC with SI-Free PAPR Reduction and Pilotless Sensing

Bensheng Yang, Min Fan, Haitao Zhao, Haiming Wang · arXiv eess.SP · 2026-06-18

★★★★■ Solid contribution addressing practical OFDM-ISAC challenges with experimental validation, though the spectral efficiency trade-off and scalability to massive MIMO or multi-user scenarios remain unclear.

This paper introduces amplitude-phase-frequency block modulation (APFBM) for OFDM-ISAC, mapping information symbols to Jones vectors on the Stokes sphere with deterministic per-block phase references. The transmitter exploits common-phase degrees of freedom for side-information-free PAPR reduction, while the receiver uses Viterbi ML detection to jointly recover optimization phases and estimate block-wise channel parameters without dedicated pilots. Over-the-air SDR measurements validate PAPR reduction and pilotless channel sensing at the cost of moderate spectral efficiency loss.

Contribution: First scheme to achieve waveform-level ISAC integration where the same deterministic phase structure simultaneously enables transmitter PAPR optimization without side information and receiver pilotless channel sensing through joint ML sequence detection.

Relevance to your work: Directly relevant to ISAC research, particularly the pilotless sensing approach which aligns with semantic/task-oriented communication principles by embedding sensing in the communication waveform structure rather than requiring dedicated overhead.

Cyclic-Prefix OFDM Probing for Spatial-ISI-Free Distributed Acoustic Sensing via Frequency-Domain Channel Reconstruction

Huan Huang, Zhiyang Xue, Ziang Chen, Zhongxing Tian, et al. · arXiv eess.SP · 2026-06-18

★★★★■ Solid incremental work demonstrating ISAC principles in the optical domain with novel signal processing, but limited direct impact on RF wireless 6G systems that are the reader's primary focus.

This paper proposes a cyclic-prefix OFDM waveform for distributed acoustic sensing (DAS) in optical fiber systems that eliminates spatial inter-symbol interference by formulating Rayleigh backscattering as a finite-memory multipath channel and applying frequency-domain equalization. The approach simultaneously performs vibration sensing over 5.2 km with sub-meter spatial resolution and error-free data communication using the same waveform. Experimental results show 29.55 dB improvement in phase reconstruction accuracy over matched-filter methods while recovering 500 Hz PZT vibrations and transmitting data without bit errors.

Contribution: First formulation of distributed Rayleigh backscattering as a finite-memory sensing channel enabling CP-OFDM frequency-domain processing to achieve deterministic spatial-ISI-free reconstruction, validated in a dual-function optical-fiber ISAC demonstration.

Relevance to your work: Tangential—demonstrates ISAC waveform sharing and channel reconstruction principles applicable across domains, but targets optical fiber DAS rather than RF air interfaces for beyond-5G wireless.

Ray Antenna Array Enhanced Low-Altitude ISAC: Performance Analysis and Beamforming Design

Zhiqiang Xiao, Tao Zhang, Hao Wu, Xiaoqiang Qiao, et al. · arXiv cs.IT · 2026-06-17

★★★★ Addresses a practical deployment challenge for ISAC systems (overhead coverage) with a novel antenna architecture, though RAA technology itself is emerging and needs further validation beyond simulations.

This paper proposes using ray antenna arrays (RAA) for low-altitude integrated sensing and communication (ISAC) systems to overcome conventional antenna limitations in overhead coverage and cost. RAA employs multiple ray-arranged arrays without phase shifters, enabling flexible beamforming through dynamic ray selection while reducing hardware complexity. An alternating optimization algorithm jointly optimizes ray selection and beamforming to maximize sensing coverage under communication constraints, demonstrating improved sensing SNR over traditional arrays.

Contribution: First application of ray antenna array architecture to low-altitude ISAC, eliminating overhead coverage holes and providing uniform angular resolution with reduced hardware cost compared to conventional phased arrays.

Relevance to your work: Directly relevant to your ISAC research interests, particularly in system-level design and beamforming strategies, though the antenna hardware focus is peripheral to AI-native PHY and learned approaches you emphasize.

Holographic Integrated Sensing and Communication With Limited Radiation Amplitudes: How Many Quantization Bits Are Enough?

Shuhao Zeng · arXiv cs.IT · 2026-06-16

★★★★ Addresses a practical implementation gap in holographic ISAC research with solid theoretical bounds, though focused on a narrow aspect (amplitude quantization) of the broader RHS deployment challenge.

The paper investigates reconfigurable holographic surfaces (RHS) for integrated sensing and communication (ISAC), focusing on practical implementation constraints. It derives closed-form lower bounds for communication rate and sensing SINR under amplitude quantization, establishing an upper bound on the minimum quantization bits required. Theoretical analysis shows how the communication-sensing tradeoff affects quantization requirements.

Contribution: First theoretical characterization of minimum amplitude quantization bits needed for holographic ISAC, providing practical design guidance for RHS implementations that previously assumed continuous amplitude control.

Relevance to your work: Directly relevant to your ISAC research interests, particularly concerning practical constraints in massive antenna systems and the fundamental tradeoffs between sensing and communication performance.

Toward Quantum-Enhanced ISAC: Active-RIS-Aided Integrated Sensing and Communication with Rydberg Atomic Receivers

Hong-Bae Jeon, Hyung-Joo Moon, Yonghwi Kim · arXiv eess.SP · 2026-06-16

★★★★ Solid theoretical contribution exploring quantum-enhanced sensing for ISAC, though Rydberg atomic receivers remain far from commercial viability and the work is primarily analytical with simulation validation.

This paper designs an active-RIS-aided ISAC system using Rydberg atomic receivers (RARE), which provide magnitude-only, real-domain observations. The authors derive a Cramér-Rao bound for direction-of-arrival estimation under RARE constraints and propose an alternating optimization framework (SDR + majorization-minimization) to jointly design BS beamforming and active-RIS reflection coefficients. Simulations show the RARE-aware design significantly outperforms conventional RF receivers and approaches radar-only performance.

Contribution: First unified ISAC model and joint optimization framework specifically tailored to the magnitude-only observation structure of quantum Rydberg atomic receivers combined with active reconfigurable intelligent surfaces.

Relevance to your work: Directly relevant to your ISAC research interests; introduces quantum sensing modality that could inform future receiver architectures, though practical deployment timeline is uncertain and current active-RIS hardware constraints may differ significantly.

Rotatable Antenna-Enhanced Secure Integrated Sensing and Communications Under Imperfect CSI

Qi Yang, Kai Liu, Jingjing Zhao, Xidong Mu, et al. · arXiv eess.SP · 2026-06-17

★★★★ Competent optimization work on a niche intersection (RAs, physical-layer security, ISAC) but limited practical impact given mechanical rotation constraints and narrow scope compared to electronic beamforming approaches dominating 6G discussion.

This paper studies a rotatable antenna (RA) system for secure ISAC where a transceiver simultaneously serves legitimate users and senses a target that may eavesdrop, under imperfect CSI of the eavesdropping channel. The authors jointly optimize transmit beamforming, artificial noise covariance, and antenna boresight angles via alternating optimization (S-procedure, SCA, SDR for beamforming/AN; particle swarm optimization for boresight). Simulations demonstrate that mechanically rotatable antennas significantly improve secrecy rates and robustness compared to fixed-antenna baselines under CSI uncertainty.

Contribution: Novel application of rotatable antennas to physical-layer security in ISAC, showing that mechanical beam steering via boresight optimization enhances secure communication against an eavesdropping sensing target under imperfect CSI.

Relevance to your work: Directly relevant to your ISAC research interests, though the rotatable antenna hardware assumption may be less applicable than your typical focus on AI/ML-based PHY and electronic beamforming for sensing-communication coexistence.

AI-native air interface

Site-Specific MIMO Channel Generation via Diffusion and Flow Matching: Fidelity, Efficiency, and Downstream Utility

Sina Beyraghi, Masoud Sadeghian, Firdous Bin Ismail, Angel Lozano, et al. · arXiv cs.IT · 2026-06-18

★★★★ Addresses a critical bottleneck in AI-native air interfaces (lack of site-specific training data) with a practical generative modeling approach validated on real propagation scenarios and downstream PHY tasks, from credible authors in the field.

This paper applies conditional diffusion (cDDIM) and flow matching (cFMM) models to generate site-specific MIMO channel matrices conditioned on user location, targeting scenarios where measurement data is scarce. Both models preserve spatial structure and achieve high statistical fidelity (beam consistency, effective rank) across 28 GHz and 3.5 GHz deployments. Downstream tasks (CSI compression, beam alignment) show substantial performance gains when augmenting limited real data with synthetic channels, with cFMM matching cDDIM quality at $\sim 10\times$ faster inference.

Contribution: First systematic comparison of diffusion and flow matching for location-conditioned MIMO channel synthesis, demonstrating that flow matching achieves comparable fidelity to diffusion with order-of-magnitude speedup and that synthetic augmentation significantly improves data-scarce downstream tasks.

Relevance to your work: Directly relevant: provides a data augmentation pathway for your neural receiver and learned PHY work, complements your interest in GAN-based channel models by offering diffusion/flow alternatives, and enables CSI compression and beam management under data scarcity.

ConsisFormer: Compute-Efficient Transformer for Wireless Foundation Models Based on Channel Consistency

Yuwei Wang, Li Sun, Tingting Yang, Liwen Jing, et al. · arXiv eess.SP · 2026-06-18

★★★★■ Addresses a critical deployment barrier (inference latency) for WFM with a principled, channel-aware efficiency solution validated across multiple 6G tasks, though real-hardware validation and comparison with state-space models would strengthen the contribution.

This paper addresses the $O(N^2)$ complexity of Transformer-based wireless foundation models by exploiting short-term channel consistency. An adaptive token aggregation module merges similar adjacent CSI tokens before self-attention, and feature sequence interpolation recovers full resolution afterward. An aggregated auto-encoder pre-training paradigm enables the model to learn robust representations from sparsified tokens, achieving 83% complexity reduction with negligible accuracy loss across channel prediction, LoS/NLOS classification, beam prediction, and localization tasks.

Contribution: First compute-efficient Transformer design specifically for wireless foundation models that exploits temporal/frequency channel correlation structure to reduce token sequence length dynamically, paired with a matching pre-training strategy for sparsified CSI representations.

Relevance to your work: Directly relevant: you are developing AI-native air interfaces and this provides a practical backbone architecture for foundation models that could support your work on neural receivers, CSI compression, beam management, and ISAC by making multi-task learning deployable under latency constraints.

6G Demos & Trials

OmniPlan: An Adaptive Framework for Timely and Near-Optimal Network Planning Optimization

Longlong Zhu, Jiashuo Yu, Zedi Chen, Yuhan Wu, et al. · arXiv cs.NI · 2026-06-16

★★★★■ Solid systems contribution for resource orchestration in edge/cloud networks, though the ML inference offloading use case is narrower than core 6G air-interface challenges and the architectural innovation is primarily in solver orchestration rather than wireless-specific optimization.

OmniPlan addresses network planning optimization (resource allocation under constraints) by combining mixed-integer programming solvers, heuristics, and deep reinforcement learning into a mixture-of-experts framework that selects solution methods based on user intent parsed by an LLM. The system dynamically balances execution time versus solution quality, demonstrated on distributed ML inference offloading across network devices. Experiments show up to 97.8% latency reduction and 11.5% resource savings compared to baseline approaches.

Contribution: Novel use of LLM-based intent parsing to dynamically select among MIP/heuristic/DRL solvers in a mixture-of-experts architecture, enabling adaptive trade-offs between planning speed and optimality for network resource allocation.

Relevance to your work: Potentially relevant for future work on intent-driven network slicing or real-time resource allocation in NTN constellations where beam assignment and computational offloading must adapt to heterogeneous traffic demands, though the paper focuses on generic network planning rather than PHY-layer or air-interface problems.

UAV-CAS: A Calibrated Digital-Twin Dataset for Intrusion Detection in UAV Swarm Networks

Sripath Mishra, Bharat Bhargava, Zizheng Liu, Shafkat Islam · arXiv cs.NI · 2026-06-16

★★■ Competent networking contribution focused on cybersecurity for UAV swarms, but outside the physical-layer AI and 6G air-interface scope central to Beyond-5G research.

The paper introduces UAV-CAS, a labeled flow dataset for intrusion detection in UAV swarm networks, generated via a ContainerNet digital twin calibrated against AERPAW testbed measurements. The four-layer calibration pipeline accounts for altitude-dependent path loss, mobility, link-level performance, and trace fidelity across 99,492 flows spanning five attack families. Binary attack detection achieves >0.98 F1, but multi-class classification remains challenging, especially for stealth attacks (F1 in single digits).

Contribution: First systematically calibrated digital-twin dataset for UAV network intrusion detection that quantifies the distribution shift between simulated swarm traffic and real testbed measurements, enabling

reproducible ML training under mobility and link-quality variations.

Relevance to your work: Tangential—addresses non-terrestrial vehicle networks but not the air-interface, channel modeling, or PHY-layer ML problems that define next-generation wireless system design.

Critical Percolation as a Synthetic Data Model for Interpretability

Aryeh Brill, Tom Ingebrechtsen Carlson · arXiv cs.LG · 2026-06-18

★☆☆☆☆ Competent ML interpretability work but narrowly focused on synthetic data generation for model analysis rather than wireless system applications.

The paper proposes using critical percolation clusters as synthetic datasets for evaluating neural network interpretability methods. The data exhibits hierarchical, multi-scale structure with sparse fractal clusters and power-law distributions, mimicking natural data properties. An efficient sampling algorithm enables arbitrary-scale generation, and probing shows ground-truth latent variables are linearly decodable from network activations.

Contribution: First principled synthetic data model with analytically tractable hierarchical structure and realistic power-law statistics for testing interpretability methods at scale.

Relevance to your work: Tangential—while neural network interpretability could inform explainable AI for learned PHY components, this work addresses ML foundations rather than wireless-specific challenges.

A Model-Driven Approach for Developing Families of Reinforcement Learning Environments

Xiaoran Liu, Istvan David · arXiv cs.LG · 2026-06-18

★☆☆☆☆ While competent software engineering work for RL environment generation, the contribution is methodological tooling rather than advancing wireless AI/ML techniques or addressing 6G physical layer challenges.

The paper proposes a model-driven approach using hybrid genetic algorithms to automatically generate families of reinforcement learning training environments, addressing the labor-intensive manual process of creating environment variants needed for RL convergence. The method expresses mutations and constraints as model transformations operationalized through a search process, demonstrated on wildfire mitigation and curriculum learning scenarios. The approach aims to enable scalable development of environment families required for realistic RL problems.

Contribution: Automated generation of RL environment families through hybrid genetic algorithms combined with model transformation engines, replacing manual environment variant creation with a systematic search-based approach.

Relevance to your work: Tangential - the RL environment generation methodology could theoretically apply to wireless simulation environments for training neural receivers or learned PHY agents, but the paper does not address wireless-specific problems or demonstrate applicability to communications systems.

Towards Modality-imbalanced Federated Graph Learning: A Data Synthesis-based Approach

Zhengyu Wu, Hongchao Qin, Xunkai Li, Zekai Chen, et al. · arXiv cs.LG · 2026-06-18

★☆☆☆☆ While technically competent for federated ML applications, this work focuses on graph-structured data with visual/textual modalities rather than wireless physical layer challenges, making it peripheral to 6G air interface research. This paper addresses modality imbalance (missing visual/textual attributes) in federated graph learning, where different clients have incomplete multimodal data at both client and node levels. The authors propose FedMGS, which synthesizes missing modality representations in latent space using prototype-guided semantic anchors and reliability-calibrated fusion, avoiding direct data reconstruction. Experiments show up to 17.41% performance gains over baselines in multimodal federated graph tasks.

Contribution: First formalization of modality-imbalanced multimodal federated graph learning as a latent semantic synthesis problem with a practical three-component solution (availability-aware encoding, prototype-guided synthesis, reliability-calibrated fusion).

Relevance to your work: Tangential - the federated learning framework could theoretically apply to distributed wireless network optimization, but the multimodal graph learning focus (vision/text) does not directly connect to PHY-layer, channel modeling, or wireless sensing problems central to your research agenda.

A Mixed-Reality Testbed for Autonomous Vehicles

H. M. Sabbir Ahmad, Ehsan Sabouni, Emrullah Celik, Zean Wan, et al. · arXiv eess.SY · 2026-06-17

★☆☆☆☆ Competent testbed infrastructure work focused on autonomous vehicles with V2X connectivity, but limited novelty in wireless communication aspects relevant to 6G physical layer or air interface research.

The paper describes a mixed-reality testbed combining physical mobile robots with high-fidelity simulation for autonomous vehicle research, supporting multi-agent scenarios with vehicular wireless connectivity. It demonstrates a control framework using Control Barrier Functions for safety-guaranteed planning and control in Connected and Autonomous Vehicles. Experiments validate the testbed's ability to bridge simulation and hardware deployment for perception, planning, and control algorithm testing.

Contribution: A hardware-in-the-loop platform integrating physical robots with photorealistic simulation and wireless V2X connectivity, enabling scalable multi-agent CAV research with safety-critical scenario testing.

Relevance to your work: Tangential - addresses V2X application layer and autonomous vehicle control rather than AI-native air interfaces, channel modeling, or 6G physical layer research that defines your core interests.

Deep Learning-Driven Inverse Design of Doherty Power Amplifiers Using Pixelated Combiners and Dual-State Impedance Synthesis

Han Zhou, Haojie Chang, David Widen, Christian Fager · arXiv eess.SP · 2026-06-16

★☆☆☆☆ Competent RF circuit design work with good experimental results, but focused narrowly on sub-3 GHz PA optimization rather than addressing core 6G physical-layer challenges at mmWave or sub-THz frequencies.

This paper presents a deep learning methodology for designing Doherty power amplifier output combiners by combining CNNs with pixelated layout representations and genetic algorithms to optimize dual-state impedance synthesis. Two GaN HEMT prototypes demonstrate >44.2 dBm saturated output power with >71.2% peak drain efficiency and 64% efficiency at 6-dB back-off across 2.6-2.8 GHz. After digital predistortion, ACLR exceeds -51.3 dBc.

Contribution: First application of deep CNNs with pixelated layout representations to automate the notoriously difficult design of three-port Doherty combiners addressing simultaneous peak and back-off impedance requirements.

Relevance to your work: Tangential—while efficient PAs matter for massive MIMO base stations and energy-efficient 6G infrastructure, this work addresses conventional amplifier circuit topology rather than AI-native air interfaces, learned PHY, or other core research areas.

Distributed General-Purpose Agent Networks: Architecture, Key Mechanisms, and Prototypes

Shengli Zhang, Deen Ma, Zibin Lin, Taotao Wang · arXiv cs.NI · 2026-06-15

★☆☆☆☆ While technically competent on distributed systems and reputation mechanisms, this work addresses LLM agent orchestration rather than wireless PHY/MAC innovation, making it peripheral to 6G air interface research.

This paper proposes a peer-to-peer architecture for distributed networks of LLM-based autonomous agents that can discover collaborators, establish trust, and execute complex tasks across heterogeneous edge/personal devices. The authors introduce a protocol adaptation layer connecting task semantics with network operations, and present mechanisms for semantic announcement propagation (bodyless gossip), verifiable identity with multi-topic reputation (BAID + MG-EigenTrust), and semantic-gradient mechanism design for open task execution. Prototype evaluations demonstrate BAID verification overhead and resilience of MG-EigenTrust reputation under collusion attacks.

Contribution: First systematic architecture for distributed general-purpose agent networks that propagates semantic intentions and cooperation constraints rather than just data, addressing unique challenges of agent collaboration that P2P overlays and conventional multi-agent systems cannot solve independently.

Relevance to your work: Tangential—the distributed edge computing aspects could inform future device-to-device semantic communication scenarios, but the core contribution is application-layer agent coordination, not physical-layer design, channel modeling, or wireless resource management relevant to AI-native air interfaces.

Sovereign Execution Brokers: Enforcing Certificate-Bound Authority in Agentic Control Planes

Jun He, Deying Yu · arXiv cs.LG · 2026-06-18

★☆☆☆☆ This is a cloud infrastructure security and governance mechanism for autonomous software agents managing IT resources, with no connection to wireless communication, physical-layer design, or radio resource management.

This paper addresses runtime authorization for autonomous agents executing infrastructure mutations by introducing a Sovereign Execution Broker (SEB) that enforces certificate-bound execution authority. SEB acts as a mandatory enforcement point between agent proposals (certified by a separate assurance boundary) and actual infrastructure API calls, verifying certificates, checking validity windows and state drift, and minting scoped execution credentials. The authors prototype SEB on AWS and Kubernetes, measuring latency overhead, revocation propagation speed, and resilience under fault injection.

Contribution: Introduction of a mandatory runtime enforcement layer that decouples agentic proposal from execution permission through short-lived, certificate-bound mutation authority with cryptographic audit trails.

Relevance to your work: tangential

Predicting gestational age at birth in the context of preterm birth from multi-modal fetal MRI

Diego Fajardo-Rojas, Megan Hall, Daniel Cromb, Mary A. Rutherford, et al. · arXiv cs.LG · 2026-06-18

★☆☆☆☆ Medical imaging application with no connection to wireless communications, physical layer design, or 6G system architecture.

This paper develops a machine learning pipeline to predict gestational age at birth from multi-modal fetal MRI data (morphological and functional), evaluated on 333 control and 93 preterm birth cases. The approach uses custom methods for imputation, feature selection, and regression, identifying cervical length and placental T2* statistics as key predictive features. The pipeline achieves $R^2 = 0.13$, MAE = 2.74 weeks, with 77% accuracy in term/preterm classification.

Contribution: First regression-based approach to predict continuous gestational age at birth rather than binary preterm classification, combining multi-modal fetal MRI with ML pipeline including motion-robust imaging.

Relevance to your work: tangential